Forces & Equilibrium

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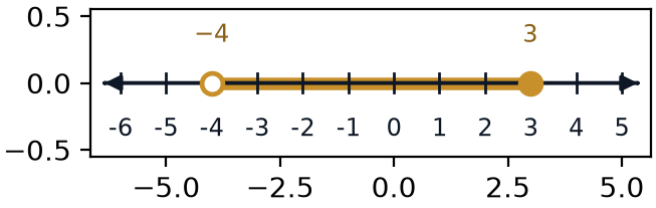
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Resolving Forces in Two Dimensions

Solve the inequality $x^2 + 17x + 72 > 0$.

2 [Foundational] • Target: Friction & Limiting Equilibrium

Solve the inequality $x^2 + x - 20 > 0$.



3 [Foundational] • Target: Inclined Planes & Equilibrium

Solve the inequality $x^2 + 7x - 8 > 0$.

4 [Foundational] • Target: Lami's Theorem & Connected Systems

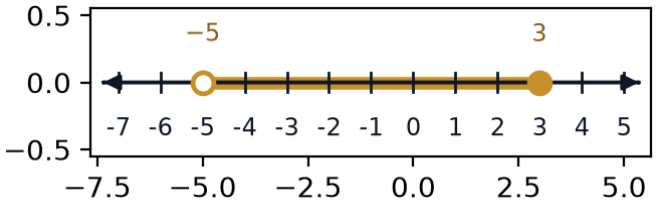
Solve the inequality $x^2 - 4x - 5 > 0$.

5 [Foundational] • Target: Resolving Forces in Two Dimensions

Solve the inequality $x^2 - 8x + 12 > 0$.

6 [Foundational] • Target: Friction & Limiting Equilibrium

Solve the inequality $x^2 + 6x > 0$.



7 [Foundational] • Target: Inclined Planes & Equilibrium

A particle of mass m kg is placed on a rough inclined plane. The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The particle is on the point of sliding down the plane. The coefficient of static friction between the particle and the plane is μ_s. Given that the acceleration due to gravity is $g = 9.8$ m/s^2, find the value of μ_s.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Lami's Theorem & Connected Systems

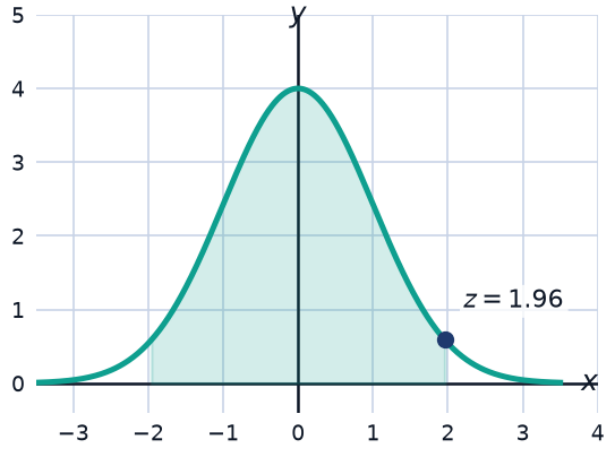
A block rests on a rough horizontal surface with normal reaction $N = 42$ N. The coefficient of friction is $\mu = 7/20$. Find the limiting frictional force.

9 [Intermediate] • Target: Resolving Forces in Two Dimensions

A block rests on a rough horizontal surface with normal reaction $N = 55$ N. The coefficient of friction is $\mu = 13/20$. Find the limiting frictional force.

10 [Intermediate] • Target: Friction & Limiting Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 16$ N. The coefficient of friction is $\mu = 11/20$. Find the limiting frictional force.



11 [Intermediate] • Target: Inclined Planes & Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 15$ N. The coefficient of friction is $\mu = 7/10$. Find the limiting frictional force.

12 [Intermediate] • Target: Lami's Theorem & Connected Systems

A block rests on a rough horizontal surface with normal reaction $N = 45$ N. The coefficient of friction is $\mu = 7/10$. Find the limiting frictional force.

13 [Intermediate] • Target: Resolving Forces in Two Dimensions

A block rests on a rough horizontal surface with normal reaction $N = 12$ N. The coefficient of friction is $\mu = 3/10$. Find the limiting frictional force.

14 [Intermediate] • Target: Friction & Limiting Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 18$ N. The coefficient of friction is $\mu = 7/10$. Find the limiting frictional force.

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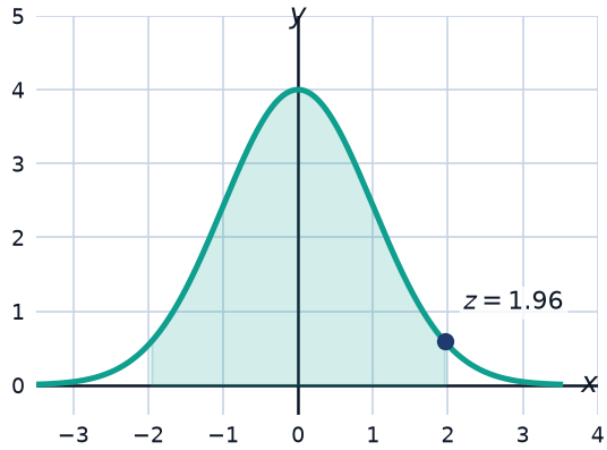
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Inclined Planes & Equilibrium • Cross-Lesson: 9.01 Circular Measure

A block rests on a rough horizontal surface with normal reaction $N = 16$ N. The coefficient of friction is $\mu = 7/20$. Find the limiting frictional force.

16 [Challenging] • Target: Lami's Theorem & Connected Systems • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A block rests on a rough horizontal surface with normal reaction $N = 55$ N. The coefficient of friction is $\mu = 9/20$. Find the limiting frictional force.



17 [Challenging] • Target: Resolving Forces in Two Dimensions • Cross-Lesson: 10.01 Series & Binomial Theorem

A block rests on a rough horizontal surface with normal reaction $N = 55$ N. The coefficient of friction is $\mu = 3/20$. Find the limiting frictional force.

18 [Challenging] • Target: Friction & Limiting Equilibrium • Cross-Lesson: 11.01 Kinematics

A block rests on a rough horizontal surface with normal reaction $N = 55$ N. The coefficient of friction is $\mu = 7/20$. Find the limiting frictional force.

19 [Challenging] • Target: Inclined Planes & Equilibrium • Cross-Lesson: 13.01 Newton's Laws

A block rests on a rough horizontal surface with normal reaction $N = 30$ N. The coefficient of friction is $\mu = 1/4$. Find the limiting frictional force.

20 [Challenging] • Target: Lami's Theorem & Connected Systems • Cross-Lesson: 14.01 Energy, Work & Power

A block rests on a rough horizontal surface with normal reaction $N = 50$ N. The coefficient of friction is $\mu = 1/4$. Find the limiting frictional force.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Resolving Forces in Two Dimensions

A force of magnitude 50 N acts at an angle of 30° above the horizontal. Find its horizontal and vertical components.

a. Horizontal: 43.3 N, Vertical: 25.0 N

b. Horizontal: 25.0 N, Vertical: 43.3 N

c. Horizontal: 35.4 N, Vertical: 35.4 N

d. Horizontal: 50.0 N, Vertical: 0.0 N

22 [Foundational] • Target: Resolving Forces in Two Dimensions

Two forces $F_1 = [8, -3]$ N and $F_2 = [-5, 7]$ N act on a particle. What is the magnitude of the resultant force?

a. 5 N

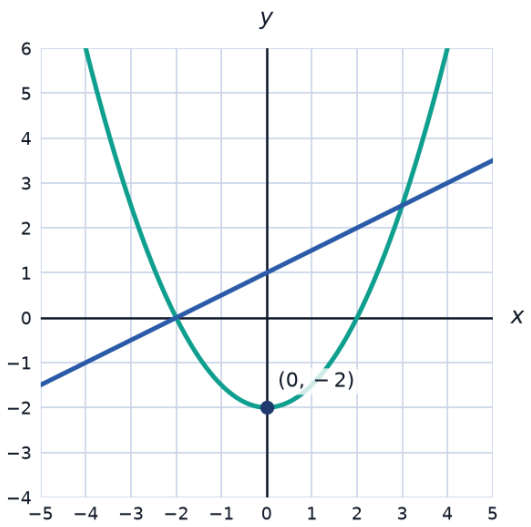
b. 7 N

c. $\sqrt{41}$ N

d. 12 N

23 [Foundational] • Target: Resolving Forces in Two Dimensions

A particle is held in equilibrium by three coplanar forces: P acting East, Q acting North, and a force of 13 N acting at an angle θ South of West where $\tan\theta = 5/12$. Find the magnitude of P.



a. 12 N

b. 5 N

c. 13 N

d. 8 N

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24 [Foundational] • Target: Friction & Limiting Equilibrium

A block of mass 6 kg rests on a rough horizontal table where the coefficient of friction is $\mu = 0.35$. Taking $g = 10$ m/s^2, what is the maximum frictional force that can act on the block?

a. 21 N

b. 60 N

c. 18 N

d. 24 N

25 [Foundational] • Target: Friction & Limiting Equilibrium

A horizontal force of 15 N is applied to the 6 kg block in the previous question (where $F_{\max} = 21$ N). What is the actual frictional force acting on the block?

a. 15 N

b. 21 N

c. 0 N

d. 6 N

26 [Foundational] • Target: Friction & Limiting Equilibrium

A particle of mass 4 kg is placed on a rough plane inclined at 30° to the horizontal. If the particle is on the point of slipping down the plane, find the coefficient of friction μ.

a. $\mu = \tan(30^\circ) = \frac{\sqrt{3}}{3} \approx 0.577$

b. $\mu = \sin(30^\circ) = 0.5$

c. $\mu = \cos(30^\circ) \approx 0.866$

d. $\mu = 0.25$

27 [Foundational] • Target: Inclined Planes & Equilibrium

A crate of mass 10 kg is held in equilibrium on a smooth slope inclined at 30° to the horizontal by a force P acting parallel to the slope. Taking $g = 10$ m/s^2, find P.

a. 50 N

b. 86.6 N

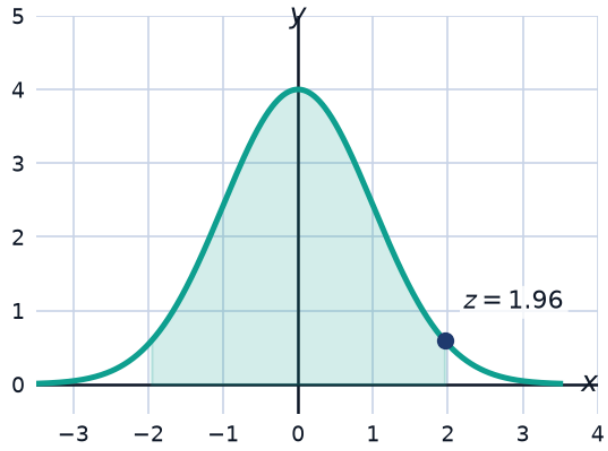
c. 100 N

d. 25 N

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28 [Foundational] • Target: Inclined Planes & Equilibrium

For the crate in the previous question, what is the normal contact force exerted by the slope on the crate?



a. 86.6 N ($50\sqrt{3}$ N)

b. 50 N

c. 100 N

d. 75 N

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Inclined Planes & Equilibrium

A body of mass 8 kg rests in limiting equilibrium on an inclined plane. The normal reaction is 64 N. Taking $g = 10$ m/s^2, find the angle of inclination α of the plane.

a. $\alpha = \arccos(0.8) \approx 36.9^\circ$

b. $\alpha = 30^\circ$

c. $\alpha = 45^\circ$

d. $\alpha = 60^\circ$

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30 [Intermediate] • Target: Lami's Theorem & Connected Systems

A lamp of weight $W = 60$ N is suspended by two light strings making angles of 30° and 60° with the horizontal ceiling. Find the tension in the steeper string.

a. $30\sqrt{3}$ N ≈ 52.0 N

b. 30 N

c. 60 N

d. 45 N

31 [Intermediate] • Target: Lami's Theorem & Connected Systems

A light inextensible string passes over a smooth peg. A mass of 5 kg hangs on one side and a mass of 3 kg hangs on the other. What downward external force must be applied to the 3 kg mass to maintain equilibrium? ($g = 10$ m/s^2)

a. 20 N

b. 50 N

c. 30 N

d. 10 N

32 [Intermediate] • Target: Lami's Theorem & Connected Systems

A ring of mass m is threaded on a rough horizontal wire with coefficient of friction $\mu = 0.4$. A light string attached to the ring is pulled with tension T at an angle of 45° above the wire. Find the value of T (in terms of mg) when the ring is in limiting equilibrium.

a. $T = \dfrac{0.4mg}{\cos(45^\circ) + 0.4\sin(45^\circ)} = \dfrac{0.4mg}{1.4\frac{\sqrt{2}}{2}} = \dfrac{4\sqrt{2}}{7}mg$

b. $\dfrac{\sqrt{2}}{2}mg$

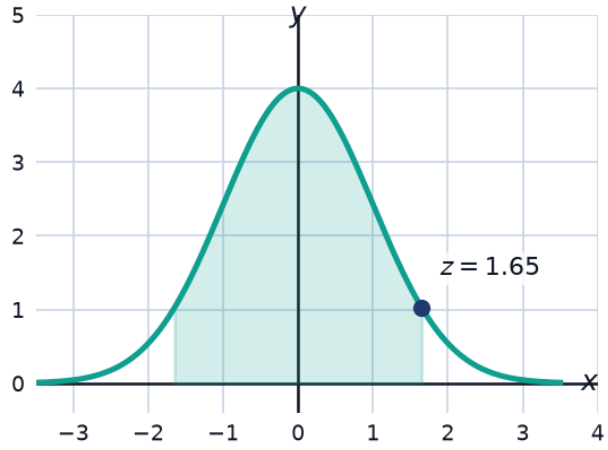
c. $0.4mg$

d. mg

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33 [Intermediate] • Target: Resolving Forces in Two Dimensions

A block rests on a rough horizontal surface with normal reaction $N = 18$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.



a. 46/5

b. 26/5

c. 72/5

d. 36/5

34 [Intermediate] • Target: Friction & Limiting Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 18$ N. The coefficient of friction is $\mu = 3/5$. Find the limiting frictional force.

a. 44/5

b. 64/5

c. 54/5

d. 108/5

35 [Intermediate] • Target: Inclined Planes & Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 15$ N. The coefficient of friction is $\mu = 1/4$. Find the limiting frictional force.

a. 7/4

b. 23/4

c. 15/2

d. 15/4

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36 [Intermediate] • Target: Lami's Theorem & Connected Systems

A block rests on a rough horizontal surface with normal reaction $N = 50$ N. The coefficient of friction is $\mu = 3/10$. Find the limiting frictional force.

a. 15

b. 17

c. 30

d. 13

37 [Intermediate] • Target: Resolving Forces in Two Dimensions

A block rests on a rough horizontal surface with normal reaction $N = 40$ N. The coefficient of friction is $\mu = 3/10$. Find the limiting frictional force.

a. 24

b. 14

c. 10

d. 12

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Friction & Limiting Equilibrium • Cross-Lesson: 11.01 Kinematics

A block rests on a rough horizontal surface with normal reaction $N = 40$ N. The coefficient of friction is $\mu = 1/2$. Find the limiting frictional force.

a. 20

b. 40

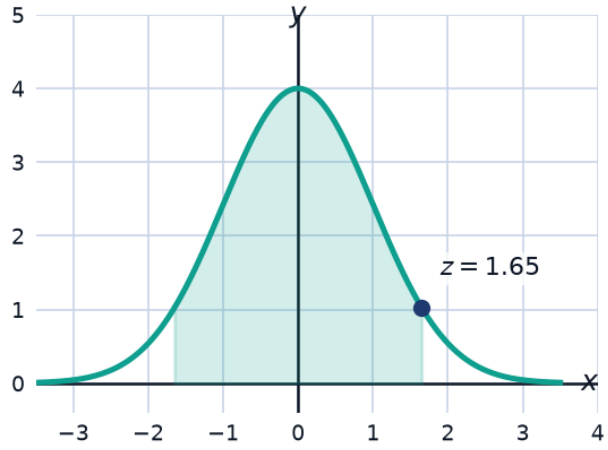
c. 18

d. 22

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39 [Challenging] • Target: Inclined Planes & Equilibrium • Cross-Lesson: 13.01 Newton's Laws

A block rests on a rough horizontal surface with normal reaction $N = 30$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.



a. 12

b. 14

c. 24

d. 10

40 [Challenging] • Target: Lami's Theorem & Connected Systems • Cross-Lesson: 14.01 Energy, Work & Power

A block rests on a rough horizontal surface with normal reaction $N = 40$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.

a. 14

b. 18

c. 32

d. 16

41 [Challenging] • Target: Resolving Forces in Two Dimensions • Cross-Lesson: 15.01 Representation of Data

A block rests on a rough horizontal surface with normal reaction $N = 45$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.

a. 16

b. 20

c. 18

d. 36

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42 [Challenging] • Target: Friction & Limiting Equilibrium • Cross-Lesson: 16.01 Permutations & Combinations

A block rests on a rough horizontal surface with normal reaction $N = 30$ N. The coefficient of friction is $\mu = 3/10$. Find the limiting frictional force.

a. 18

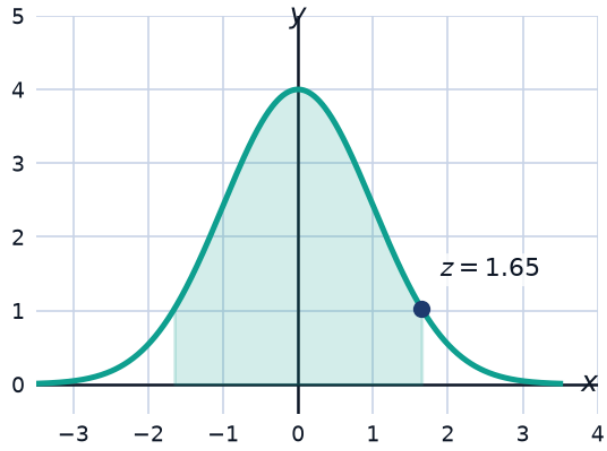
b. 7

c. 9

d. 11

43 [Challenging] • Target: Inclined Planes & Equilibrium • Cross-Lesson: 17.01 Discrete Random Variables

A block rests on a rough horizontal surface with normal reaction $N = 10$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.



a. 4

b. 8

c. 6

d. 2

44 [Challenging] • Target: Lami's Theorem & Connected Systems • Cross-Lesson: 18.01 Binomial Distribution

A block rests on a rough horizontal surface with normal reaction $N = 20$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.

a. 6

b. 16

c. 10

d. 8

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45 [Challenging] • Target: Resolving Forces in Two Dimensions • Cross-Lesson: 19.01 Normal Distribution

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 20 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

a. 18

b. 20

c. 25

d. 22

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